Mir Ahnaf Ali

HS 380

Annotated Bibliography

[1] Reddit, "Opening front door with Apple Pay," *r/homeautomation*, 2022. [Online]. Available: https://www.reddit.com/r/homeautomation/comments/tgvr7z/opening_front_door_with_apple_pay/ (accessed Nov. 11, 2025).
Why used: real-world Arduino + NFC workflow and constraints when trying to treat Apple Pay/Watch as a "key." Useful for reliability expectations, UX flow, and what breaks in practice—good material for a Design Considerations section and risk notes. (Reddit)

[2] Apple, "NFCVASReaderSession," *Apple Developer Documentation*. [Online]. Available: https://developer.apple.com/documentation/corenfc/nfcvasreadersession (accessed Nov. 11, 2025).
Why used: the official entry point for **Apple VAS** (non-payment passes like IDs/loyalty). Clarifies entitlement requirements, session lifecycle, and data flow—exactly what I need if I pivot from raw EMV scraping to a wallet-friendly, standards-aligned reader path. (Apple Developer)

[3] Apple, "Contactless passes in Apple Pay," *Apple Platform Security*. [Online]. Available: https://support.apple.com/guide/security/contactless-passes-in-apple-pay-secbd55491ad/web (accessed Nov. 11, 2025).
Why used: production-level description of VAS behavior (short range, independent vs. co-present with Apple Pay). Helps me argue security and UX trade-offs in an attendance context and explain why a pass-based approach is cleaner than payment emulation. (Apple Support)

[4] EMVCo, "EMV® Contactless Chip," *EMVCo.com*. [Online]. Available: https://www.emvco.com/emv-technologies/emv-contactless-chip/ (accessed Nov. 11, 2025).
Why used: baseline on how **EMV contactless** actually works (cryptograms, kernels, approvals). I cite this to show why a PN532 hobby reader is **not** an EMV terminal and to separate payment rails from NFC Forum/ID use cases in the thesis background. (EMVCo)

[5] T.-W. Chiang, C.-S. Wang, and C.-C. Chen, "Development and evaluation of an attendance tracking system using Android smartphones with GPS and NFC," *Applied Artificial Intelligence*, vol. 36, no. 1, 2022. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/08839514.2022.2083796 (accessed Nov. 11, 2025).

Why used: **peer-reviewed** implementation/evaluation using NFC on phones; offers metrics, methodology, and validation patterns to adapt for your thesis evaluation.

[6] NFC Forum, “Data Exchange Format (NDEF) Technical Specification,” *nfc-forum.org*. [Online]. Available: https://nfc-forum.org/build/specifications/data-exchange-format-ndef-technical-specification/ (accessed Nov. 11, 2025).
Why used: the payload model I need if I support **phone-as-tag**, encoded IDs, or app/URL handoffs. Anchors the interoperability story and future-proofs the data layer. (NFC Forum)

[7] Adafruit, “Overview—PN532 RFID/NFC Breakout and Shield,” *Adafruit Learn*. [Online]. Available: https://learn.adafruit.com/adafruit-pn532-rfid-nfc/overview (accessed Nov. 11, 2025).
Why used: practical notes for Methods—wiring (I^2C/SPI/UART), 3.3 V logic, antenna placement, supported tag families. This is the community-standard quickstart I can reference for reproducibility. (Adafruit Learning System)

[8] *bleak* documentation, “Bleak—A cross-platform Bluetooth Low Energy client,” *Read the Docs*. [Online]. Available: https://bleak.readthedocs.io/ (accessed Nov. 11, 2025).
Why used: matches my Python BLE layer for streaming UIDs/events. Covers scanning, connects, notifications, and async patterns—useful to justify library choice and document the transport path. (Bleak)